

Instructions for Extended Match Questions

Use this slide to understand how to answer an EMQ

Instructions:

1. For each question, enter all the options which you find is the answer to that question as "**comma separated values**".
2. Enter the options in **ascending order** of option number.
3. An option can also be an answer to more than one question, or may not be an answer to any of the questions.
4. [Please click here to view a sample template file for answering.](#)
5. Please note that if you give an incorrect answer there will be a negative marking for the incorrect option.

List of Options:

1. T is one to one.
2. T is onto.
3. T is an isomorphism.
4. $\text{Nullspace}(T) = \{(t, t, t) \mid t \in \mathbb{R}\}$
5. $\text{Nullspace}(T) = \{(t, t, s) \mid t, s \in \mathbb{R}\}$
6. $\text{Nullspace}(T) = \{(0, 0, 0)\}$
7. A basis of $\text{Nullspace}(T) = \{(1, 1, 1)\}$
8. A basis of $\text{Nullspace}(T) = \{(1, 1, 0), (0, 0, 1)\}$
9. $\text{nullity}(T) = 1$
10. $\text{nullity}(T) = 2$

List of Questions:

Q1 Consider the linear transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by $T(x, y, z) = (x - y, y - z, 0)$. Which of the above options are true for T ?

Q2 Consider the linear transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ defined by $T(x, y, z) = (x - y, y - z)$. Which of the above options are true for T ?

Q3 Consider the linear transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by $T(x, y, z) = (x - y, y - z, z - x)$. Which of the above options are true for T ?

Q4 Consider the linear transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by $T(x, y, z) = (x - y, 0, 0)$. Which of the above options are true for T ?

Q5 Consider the linear transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by $T(x, y, z) = (x, x + y, x + y + z)$. Which of the above options are true for T ?